

EXERCISE 4.5

Find the lines represented by each of the following and also find measure of the angle between them (problems 1-6):

Q.1 $10x^2 - 23xy - 5y^2 = 0$

Solution:

$$10x^2 - 23xy - 5y^2 = 0$$

$$10x^2 - 25xy + 2xy - 5y^2 = 0$$

$$5x(2x - 5y) + y(2x - 5y) = 0$$

$$(2x - 5y)(5x + y) = 0$$

$$2x - 5y = 0, 5x + y = 0$$

$$10x^2 - 23xy - 5y^2 = 0$$

Here, $a = 10, h = -\frac{23}{2}, b = -5$

$$\text{As } \tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan \theta = \frac{2\sqrt{\left(-\frac{23}{2}\right)^2 - (10)(-5)}}{10 + (-5)}$$

$$\tan \theta = \frac{2\sqrt{\frac{529}{4} + 50}}{5}$$

$$\tan \theta = \frac{2\sqrt{\frac{529 + 200}{4}}}{5}$$

$$\tan \theta = \frac{2\sqrt{\frac{729}{4}}}{5}$$

$$\tan \theta = \frac{2 \cdot \frac{27}{2}}{5}$$

$$\tan \theta = \frac{27}{5}$$

$$\theta = \tan^{-1}\left(\frac{27}{5}\right)$$

$$\boxed{\theta = 79.51^\circ}$$

Q.2 $3x^2 + 7xy + 2y^2 = 0$

Solution:

$$3x^2 + 7xy + 2y^2 = 0$$

$$3x^2 + 6xy + xy + 2y^2 = 0$$

$$3x(x + 2y) + y(x + 2y) = 0$$

$$(x + 2y)(3x + y) = 0$$

$$x + 2y = 0, 3x + y = 0$$

$$3x^2 + 7xy + 2y^2 = 0$$

Here $a = 3, h = \frac{7}{2}, b = 2$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan \theta = \frac{2\sqrt{\left(\frac{7}{2}\right)^2 - (3)(2)}}{3 + 2}$$

$$\tan \theta = \frac{2\sqrt{\frac{49}{4} - 6}}{5}$$

$$\tan \theta = \frac{2\sqrt{\frac{49 - 24}{4}}}{5}$$

$$\tan \theta = \frac{2\sqrt{\frac{25}{4}}}{5}$$

$$\tan \theta = \frac{2 \cdot \frac{5}{2}}{5}$$

$$\tan \theta = \frac{5}{5}$$

$$\tan \theta = 1$$

$$\theta = \tan^{-1}(1)$$

$$\boxed{\theta = 45^\circ}$$

Q.3 $9x^2 + 24xy + 16y^2 = 0$

Solution:

$$9x^2 + 24xy + 16y^2 = 0$$

$$9x^2 + 12xy + 12xy + 16y^2 = 0$$

$$3x(3x + 4y) + 4y(3x + 4y) = 0$$

$$(3x + 4y)(3x + 4y) = 0$$

$$3x + 4y = 0, 3x + 4y = 0$$

As the lines are real and coincident
so angle between the lines is 0° .

Q.4 $2x^2 + 3xy - 5y^2 = 0$

Solution:

$$2x^2 + 3xy - 5y^2 = 0$$

$$2x^2 + 5xy - 2xy - 5y^2 = 0$$

$$x(2x + 5y) - y(2x + 5y) = 0$$

$$(2x + 5y)(x - y) = 0$$

$$2x + 5y = 0, \quad x - y = 0$$

$$2x^2 + 3xy - 5y^2 = 0$$

Here $a = 2, h = \frac{3}{2}, b = -5$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan \theta = \frac{2\sqrt{\left(\frac{3}{2}\right)^2 - (2)(-5)}}{2 + (-5)}$$

$$\tan \theta = \frac{2\sqrt{\frac{9}{4} + 10}}{-3}$$

$$\tan \theta = \frac{2\sqrt{\frac{9 + 40}{4}}}{-3}$$

$$\tan \theta = \frac{2\sqrt{\frac{49}{4}}}{-3}$$

$$\tan \theta = \frac{2\left(\frac{7}{2}\right)}{-3}$$

$$\tan \theta = -\frac{7}{3}$$

$$\theta = \tan^{-1}\left(-\frac{7}{3}\right)$$

$$\theta = 180^\circ - \tan^{-1}\left(\frac{7}{3}\right)$$

$$\boxed{\theta = 113.20^\circ}$$

Q.5 $6x^2 - 19xy + 15y^2 = 0$

Solution:

$$6x^2 - 19xy + 15y^2 = 0$$

$$6x^2 - 10xy - 9xy + 15y^2 = 0$$

$$2x(3x - 5y) - 3y(3x - 5y) = 0$$

$$(3x - 5y)(2x - 3y) = 0$$

$$3x - 5y = 0, \quad 2x - 3y = 0$$

$$6x^2 - 19xy + 15y^2 = 0$$

Here $a = 6, h = \frac{-19}{2}, b = 15$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan \theta = \frac{2\sqrt{\left(\frac{-19}{2}\right)^2 - (6)(15)}}{6 + 15}$$

$$\tan \theta = \frac{2\sqrt{\frac{361}{4} - 90}}{21}$$

$$\tan \theta = \frac{2\sqrt{\frac{361 - 360}{4}}}{21}$$

$$\tan \theta = \frac{2\sqrt{\frac{1}{4}}}{21}$$

$$\tan \theta = \frac{2 \cdot \frac{1}{2}}{21}$$

$$\theta = \tan^{-1}\left(\frac{1}{21}\right)$$

$$\boxed{\theta = 2.73^\circ}$$

Q.6 $x^2 + 2xy \sec \alpha + y^2 = 0$

Solution:

$$x^2 + 2xy \sec \alpha + y^2 = 0$$

Dividing both sides by x^2

$$1 + 2 \sec \alpha \left(\frac{y}{x}\right) + \frac{y^2}{x^2} = 0$$

Let $\frac{y}{x} = t \dots (i)$

$$1 + (2 \sec \alpha)t + t^2 = 0$$

$$\Rightarrow t^2 + (2 \sec \alpha)t + 1 = 0$$

Here $a = 1$, $b = 2 \sec \alpha$, $c = 1$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-2 \sec \alpha \pm \sqrt{(2 \sec \alpha)^2 - 4(1)(1)}}{2(1)}$$

$$t = \frac{-2 \sec \alpha \pm \sqrt{4 \sec^2 \alpha - 4}}{2(1)}$$

$$t = \frac{-2 \sec \alpha \pm \sqrt{4(\sec^2 \alpha - 1)}}{2}$$

$$t = \frac{-2 \sec \alpha \pm \sqrt{4 \tan^2 \alpha}}{2}$$

$$t = \frac{-2 \sec \alpha \pm 2 \tan \alpha}{2}$$

$$t = \frac{-2(-\sec \alpha \pm \tan \alpha)}{2}$$

$$t = -(\sec \alpha \pm \tan \alpha)$$

$$t = -(\sec \alpha + \tan \alpha), t = -(\sec \alpha - \tan \alpha)$$

From equation (i)

$$\frac{y}{x} = -(\sec \alpha + \tan \alpha), \frac{y}{x} = -(\sec \alpha - \tan \alpha)$$

$$y = -(\sec \alpha + \tan \alpha)x, y = -(\sec \alpha - \tan \alpha)x$$

$$(\sec \alpha + \tan \alpha)x + y = 0, (\sec \alpha - \tan \alpha)x + y = 0$$

$$x^2 + 2xy \sec \alpha + y^2 = 0$$

Here $a = 1$, $h = \sec \alpha$, $b = 1$

$$\text{As } \tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan \theta = \frac{2\sqrt{\sec^2 \alpha - (1)(1)}}{1 + 1}$$

$$\tan \theta = \frac{\cancel{2}\sqrt{\sec^2 \alpha - 1}}{\cancel{2}}$$

$$\tan \theta = \tan \alpha$$

$$\boxed{\theta = \alpha}$$

Q.7 Find a joint equation of the lines through the origin and perpendicular to the lines:

$$x^2 - 2xy \tan \alpha - y^2 = 0$$

Solution:

$$x^2 - 2xy \tan \alpha - y^2 = 0 \text{ --- (i)}$$

let $y = m_1x$, $y = m_2x$ be the lines

represented by equation (i)

Equations of lines passing through origin and perpendicular to the lines represented by equation (i) are

$$y = \frac{-1}{m_1}x, \quad y = \frac{-1}{m_2}x$$

$$m_1y = -x, \quad m_2y = -x$$

$$x + m_1y = 0, \quad x + m_2y = 0$$

Now, joint equation of these lines is

$$(x + m_1y)(x + m_2y) = 0$$

$$x^2 + m_1xy + m_2xy + m_1m_2y^2 = 0$$

$$x^2 + (m_1 + m_2)xy + m_1m_2y^2 = 0 \text{ --- (ii)}$$

From equation (i)

$$x^2 - 2xy \tan \alpha - y^2 = 0$$

Here $a = 1, h = -\tan \alpha, b = -1$

$$m_1 + m_2 = \frac{-2h}{b} = \frac{-2(-\tan \alpha)}{-1} = -2 \tan \alpha$$

$$m_1 m_2 = \frac{a}{b} = \frac{1}{-1} = -1$$

Putting in equation (ii)

$$x^2 + (-2 \tan \alpha)xy + (-1)y^2 = 0$$

$$\boxed{x^2 - 2 \tan \alpha xy - y^2 = 0}$$

Q.8 Find a joint equation of the lines through the origin and perpendicular to the lines

$$ax^2 + 2hxy + by^2 = 0$$

Solution:

$$ax^2 + 2hxy + by^2 = 0 \dots (i)$$

Let $y = m_1 x$, $y = m_2 x$ be the lines

represented by equation (i)

Equations of lines through origin and perpendicular to the lines represented by equation (i) are

$$y = \frac{-1}{m_1} x, \quad y = \frac{-1}{m_2} x$$

$$m_1 y = -x, \quad m_2 y = -x$$

$$x + m_1 y = 0, \quad x + m_2 y = 0$$

Now, joint equation of these lines is

$$(x + m_1 y)(x + m_2 y) = 0$$

$$x^2 + m_1 xy + m_2 xy + m_1 m_2 y^2 = 0$$

$$x^2 + (m_1 + m_2)xy + m_1 m_2 y^2 = 0 \dots (ii)$$

From equation (i)

$$ax^2 + 2hxy + by^2 = 0$$

$$m_1 + m_2 = \frac{-2h}{b}$$

$$m_1 m_2 = \frac{a}{b}$$

Putting in equation (ii)

$$x^2 + \left(\frac{-2h}{b}\right)xy + \frac{a}{b}y^2 = 0$$

Multiplying by b on both sides.

$$\boxed{bx^2 - 2hxy + ay^2 = 0}$$

Q.9 Find the area of the region

$$\text{bounded by } 10x^2 - xy - 21y^2 = 0$$

$$\text{and } x + y + 1 = 0$$

Solution:

$$10x^2 - xy - 21y^2 = 0$$

$$10x^2 - 15xy + 14xy - 21y^2 = 0$$

$$5x(2x - 3y) + 7y(2x - 3y) = 0$$

$$(2x - 3y)(5x + 7y) = 0$$

$$\text{Either } 2x - 3y = 0 \text{ or } 5x + 7y = 0$$

$$2x - 3y = 0 \dots (i)$$

$$5x + 7y = 0 \dots (ii)$$

$$x + y + 1 = 0 \dots (iii)$$

Solving (i) and (iii)

By Eq (i) -2 Eq (iii)

$$2x - 3y = 0$$

$$\pm 2x \pm 2y \pm 2 = 0$$

$$\hline -5y - 2 = 0$$

$$-5y = 2$$

$$y = \frac{-2}{5}$$

Equation (iii)

$$x + y + 1 = 0$$

$$x + \left(\frac{-2}{5}\right) + 1 = 0$$

$$x = -\frac{3}{5}$$

So $A\left(\frac{-3}{5}, \frac{-2}{5}\right)$ is the point of

intersection of (i) and (iii)

By eq (ii) -5 eq (iii)

$$5x + 7y = 0$$

$$\pm 5x \pm 5y \pm 5 = 0$$

$$2y - 5 = 0$$

$$2y = 5$$

$$y = \frac{5}{2}$$

Equation (iii)

$$x + y + 1 = 0$$

$$x + \frac{5}{2} + 1 = 0$$

$$x = -\frac{7}{2}$$

So $B\left(\frac{-7}{2}, \frac{5}{2}\right)$ is the point of

intersection of (ii) and (iii)

Clearly, the point of intersection of (i) and (ii) is origin, $O(0,0)$

$$\text{Then area of } \Delta OAB = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ \frac{-3}{5} & \frac{-2}{5} & 1 \\ \frac{-7}{2} & \frac{5}{2} & 1 \end{vmatrix}$$

Expanding by R_1

$$= \frac{1}{2} \left[0 - 0 + 1 \begin{vmatrix} \frac{-3}{5} & \frac{-2}{5} \\ \frac{-7}{2} & \frac{5}{2} \end{vmatrix} \right]$$

$$= \frac{1}{2} \left[-\frac{15}{10} - \frac{14}{10} \right]$$

$$= \frac{1}{2} \left[\frac{-15-14}{10} \right]$$

$$= \frac{-29}{20}$$

As area can never be negative, so

$$\text{Area of } \Delta OAB = \frac{29}{20} \text{ square units}$$